

A Dataset Is a Collection of Observations

A dataset is an organized collection of observations.



CORE CONCEPT

NumPy Arrays Store Shaped Numerical Data

A NumPy array is a collection of values arranged along one or more axes. Array values normally share one data type. For a two-dimensional array, shape (2, 3) means 2 rows and 3 columns. Its size is $2 \times 3 = 6$.

ndim: number of axes

shape: length of each axis

size: total number of values

dtype: type used to store the values

Axes: 2

Shape: (2, 3)

RUN IT AND INSPECT THE RESULT

Indexing Selects Values from an Array

```
print(scores[1, 2])  
print(scores[:, 1])
```

OUTPUT

```
88  
[80 79]
```

`scores[1, 2]` gives 88. `scores[:, 1]` gives [80, 79].

REASONING SEQUENCE

Vectorized Operations Work on Whole Arrays

Frame

A vectorized operation applies one operation to an entire array.

1

Transform

adds 5 to every score.

2

Validate

NumPy performs the repeated numerical work in optimized array code.

3

Explain

Broadcasting is NumPy's rule for combining arrays with compatible shapes.

4

Original: [72, 84, 91] Adjustment: + 5 Result: [77, 89, 96] The same addition is applied to all three positions.

CORE CONCEPT

Series and DataFrames Add Labels

A Pandas Series is a one-dimensional labelled collection. A DataFrame is a two-dimensional labelled table. Labels make data easier to interpret:

row labels form the index;

column labels identify variables;

each column has a data type;

different columns may use different types.

RUN IT AND INSPECT THE RESULT

Load Data, Then Inspect Before Transforming

```
df = pd.read_csv("results.csv")
print(df.shape)
print(df.dtypes)
print(df.isna().sum())
```

OUTPUT

```
(120, 4)
student    object
course     object
score      float64
attendance  int64
```

If `df.shape` is `(120, 4)`, the table has 120 rows and 4 columns. This does not prove that all rows are valid; it only states the current shape.

CORE CONCEPT

Select, Filter, and Sort Rows Deliberately

Selection chooses columns or rows. Filtering keeps rows that satisfy a condition. Sorting changes row order without changing the values. The comparison creates a boolean mask: one True or False value per row. Pandas keeps rows where the mask is True. Use `.loc[rows, columns]` when selecting by labels:

Selection chooses columns or rows.

Filtering keeps rows that satisfy a condition.

Sorting changes row order without changing the values.

The comparison creates a boolean mask: one True or False value per row.

Pandas keeps rows where the mask is True.

Use `.loc[rows, columns]` when selecting by labels:

REASONING SEQUENCE

Missing Values Need an Explicit Policy

Frame

the student did not
take the test;

1

Transform

the score was not
recorded;

2

Validate

the file was parsed
incorrectly; or

3

Explain

the value was
removed by a rule.

4

Scores: [72, missing, 84] The policy changes the conclusion.

RUN IT AND INSPECT THE RESULT

Grouping Answers Questions About Categories

```
means = df.groupby("course")["score"].mean()
print(means)
```

OUTPUT

```
course
AI      91.0
Python  78.0
```

Python mean: $(72 + 84) / 2 = 78$ AI mean: $91 / 1 = 91$

QUALITY GATE

A Complete Table Analysis Has Visible Steps



load the source;



inspect shape, columns, types, and missingness;



state the row filter;



group at the intended level;

RISK REVIEW

Common Mistakes Change the Meaning

01 reading shape (3, 4) as four rows and three columns;

02 confusing a column label with a row position;

03 filtering with a mask of the wrong length;

GUIDED LAB

Guided Lab: Explain a Small Dataset

01

State what one row represents

02

Report shape, columns, and data types

03

Count missing values

04

Keep scores from 0 to 100